

D2 CSE E1

Group – G 11

T1 - Subodh Rana

T2 - Vikram Kumar

T3 – Tushar Sharma

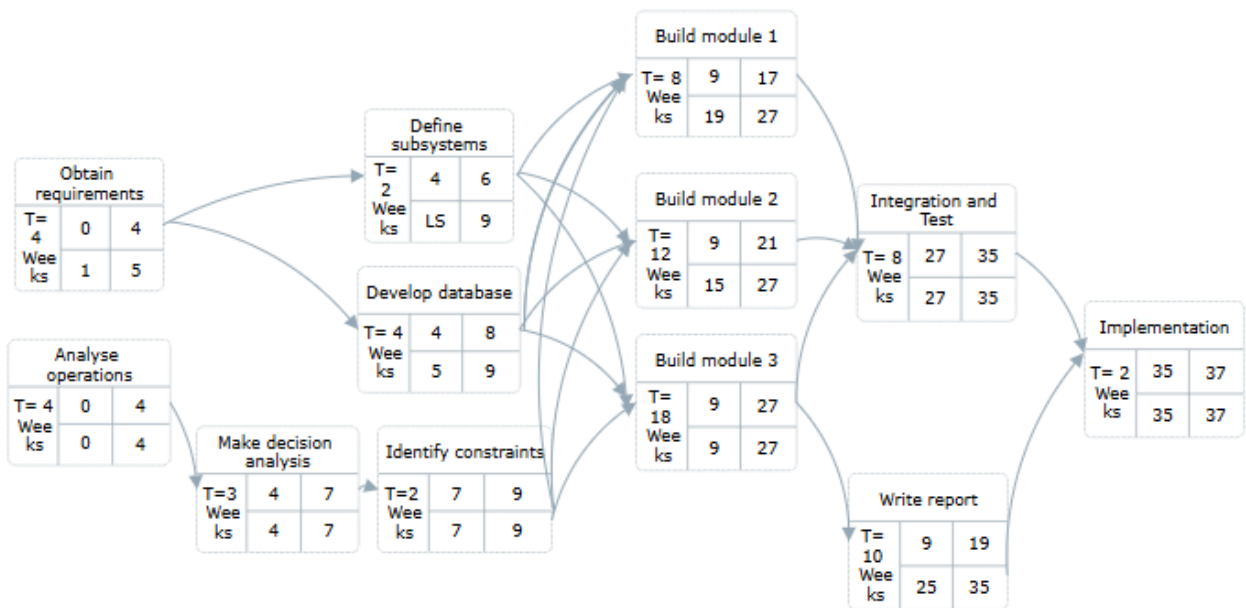
Tutorial Sheet – 3

Using any project management software of your choice (Smartdraw, edraw, openproj, lucid charts etc.)

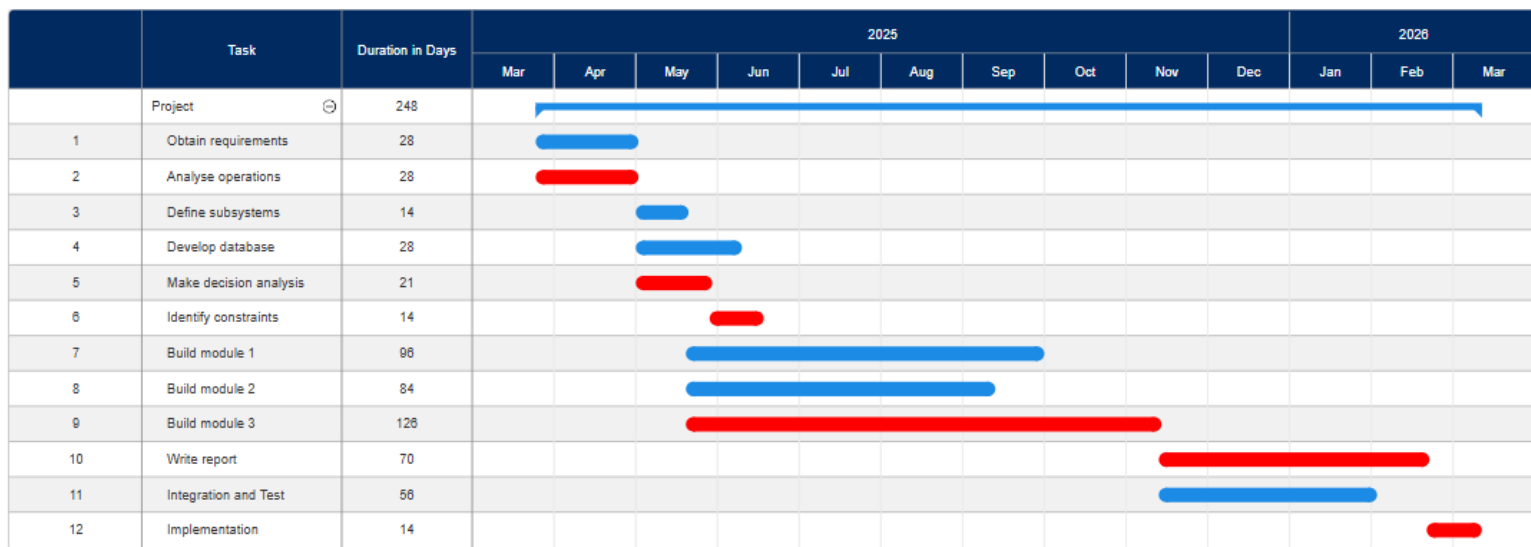
Q1. Suppose for the project selected, you are the project manager requiring the following activities

Activity No.	Activity Name	Duration (weeks)	Immediate Predecessor
1.	Obtain requirements	4	-
2.	Analyse operations	4	-
3.	Define subsystems	2	1
4.	Develop database	4	1
5.	Make decision analysis	3	2
6.	Identify constraints	2	5
7.	Build module 1	8	3,4,6
8.	Build module 2	12	3,4,6
9.	Build module 3	18	3,4,6
10.	Write report	10	6
11.	Integration and Test	8	7,8,9
12.	Implementation	2	10,11

- Draw the Activity Network representation of the project.
- Determine ES, EF, and LS, LF for every task.



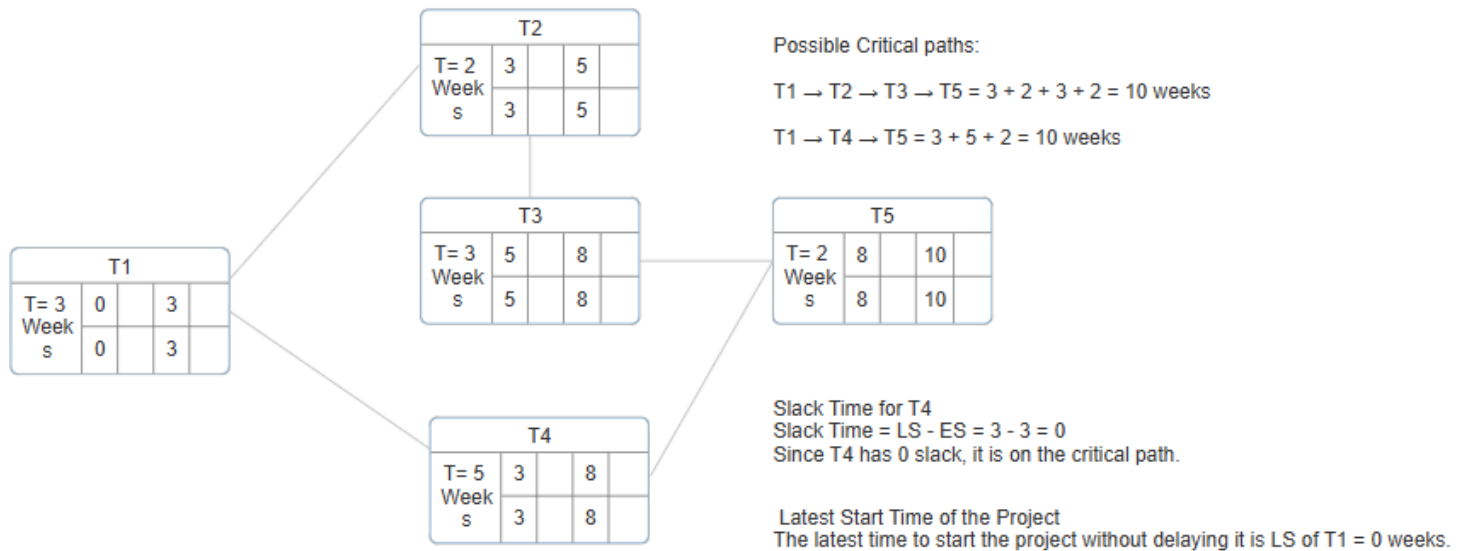
- Draw the Gantt Chart of the project.



Q2. Consider your software project with 5 tasks T1-T5. The duration of the 5 tasks in weeks are 3, 2, 3, 5, 2 respectively. T2 and T4 can start when T1 is complete. T3 can start when T2 is complete. T5 can start when both T3 and T4 are complete.

- Draw the PERT chart representation of the project.
- When is the latest Start time of the project?
- What is the slack time of task T4?
- Which tasks are on the critical path?

Task	Duration (weeks)	Immediate Predecessor
T1	3	-
T2	2	T1
T3	3	T2
T4	5	T1
T5	2	T3, T4



Q3. Suppose that you selected software product costs Rs. 50,000 to buy off-shelf and that its size is 40 KLOC. Assuming that in-house developers cost Rs. 6000 per programmer-month, would it be cost-effective to buy the product or build it?

1. Estimating Effort Required Using COCOMO Model

A widely used model for estimating software effort is the **Basic COCOMO Model**, which is given by:

$$\text{Effort}(E) = a \times (\text{KLOC})^b$$

For **Organic Projects** (small teams, familiar requirements):

$$E = 2.4 \times (\text{KLOC})^{1.05}$$

For **Semi-Detached Projects** (medium complexity):

$$E = 3.0 \times (\text{KLOC})^{1.12}$$

For **Embedded Projects** (high complexity):

$$E = 3.6 \times (KLOC)^{1.20}$$

Since we are dealing with an off-the-shelf product, we assume it is an **Organic** project.

$$E = 2.4 \times (40)^{1.05} = 115.44 \text{ person-months}$$

2. Calculating Development Cost

Given that the cost per programmer-month is ₹6000, the total development cost is:

$$\text{Total Cost} = \text{Effort} \times \text{Cost per Programmer-Month}$$

$$= 115.44 \times 6000 = ₹ 692,669$$

The estimated cost to develop the software in-house is approximately **₹692,669**.

Decision: Buy vs. Build

- **Buying Cost:** ₹50,000
- **Building Cost:** ₹692,669

Since the **in-house development cost is significantly higher than the purchase cost**, it is **far more cost-effective to buy** the software rather than build it in-house

Q4. Suppose the software product selected by you is of organic type. You have estimated the size of the product to be 100,000 lines of code. Compute the nominal effort and development time.

For an **Organic** type of software project, we use the **Basic COCOMO Model**:

$$\begin{aligned} \text{Effort}(E) &= 2.4 \times (KLOC)^{1.05} \\ \text{Development Time}(T) &= 2.5 \times (\text{Effort})^{0.38} \end{aligned}$$

where:

KLOC = 100 (since 100,000 lines of code = 100 KLOC)

a = 2.4 (Effort coefficient for Organic projects)

b = 1.05 (Effort exponent for Organic projects)

c = 2.5 (Time coefficient for Organic projects)

d = 0.38 (Time exponent for Organic projects)

$$\text{Nominal Effort} = 2.4 \times (100)^{1.05} = 302.14 \text{ person-months}$$

$$\text{Development Time} = \text{Effort} \times \text{Nominal Effort} = 2.5 \times 302.14 = 21.9 \text{ months}$$

The estimated **Nominal Effort** required to develop the software is **302.14 person-months**, and the **Development Time** is approximately **21.9 months**.